

Crowd Supervision and Vocabulary-Mismatched Selective Facial-Expression Prediction

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Abstract

Purpose: Crowd-vote supervision can alter confidence ranking without consistently improving category choice. We examine this distinction when a facial-expression predictor omits categories in the annotation vocabulary. **Methods:** Fifteen matched Swin-T runs compare hard, tie-aware hard, dominant-confidence-preserving uniform-minority, tie-preserving uniform-minority, and full soft targets on FER2013 images with FER+ votes. Exact-pixel-isolated selection, calibration and testing support common-coverage, fixed-image and fixed-epoch comparisons. Complete-vote disagreement is separated into unsupported mass, supported ambiguity and category-choice excess. Three adaptive phases reuse one image source; they are not independent confirmation. **Results:** At 80% coverage, selected-model Soft-minus-Uniform disagreement is -0.38 percentage points; Soft-minus-Tie-Uniform is -0.59. On Uniform-selected images, Soft instead differs by +0.09 points. Changing the reference on identical majority-subset images substantially changes F1. Soft retains more zero-supported-mass images than Uniform. All 36 nonempty empirical test operating points for the original four conditions exceed their calibration targets. **Conclusion:** Target representation, accepted-image composition, categorical reference and vocabulary mismatch jointly shape apparent reliability. The observed within-source contrasts support diagnostic reporting, not a novel architecture, universal superiority, deployment risk guarantee or validated messaging benefit.

Keywords: facial-expression recognition, crowd annotations, selective prediction, label distributions, supported vote mass, confidence calibration

1 Introduction

Crowd annotations make facial-expression recognition a distributional problem: people can assign different categories to the same image. A categorical reference conceals that disagreement, while a confidence-based selector can retain less ambiguous images without improving category choice. These issues become coupled when the predictor’s output vocabulary omits categories included in the annotation panel. A low selected error against dominant labels need not imply low disagreement with the complete vote distribution.

Our sender-controlled expression-cue prototype motivates this measurement problem but is not its validation setting. Related applications include EmotionPush’s text-based notifications [1], EmoChat’s facial and textual signals [2], and expression-informed emoji selection [3]. Before claiming any benefit during messaging, it is useful to establish what benchmark confidence and annotation agreement can support. Here we study that narrower reliability question.

A facial-expression category is not a direct observation of internal emotion, intention, or sincerity. Context, culture, individual differences, and deliberate expression constrain that inference [4]. Accordingly, the application suggests a fallible visual category, requires a per-message choice, and does not infer what the text means. The benchmark analysis concerns consistency with image annotations, not correctness about a person’s feelings.

We ask three questions. RQ1: how do detailed minority allocation and tie-preserving dominant-confidence softening compare under matched training? RQ2: do own-set risk differences persist when predictions are compared on identical accepted images, and how do label reference and unsupported votes affect interpretation? RQ3: what do calibration/test distribution diagnostics, calibration half-splits, and a conservative threshold comparator reveal about empirical target overshoot?

The contribution is a bounded empirical investigation of these alternative explanations, not a new architecture, loss principle, or decomposition theorem. Fifteen matched training runs, selected versus fixed-epoch evaluation, and sample-level reliability diagnostics connect supervision choices to the meaning of the selected risk. The local messaging prototype supplies motivation and an implementation artifact, not application evidence. The initial six hard/soft runs preceded review; six Uniform/Tie-hard control runs followed their known outcomes; three Tie-Uniform runs followed the second review. Each adaptive phase was specified before its training and frozen before its calibration/test inference. All analyses reuse the same image source and are not independent confirmation.

2 Related work

FER2013 supplies seven expression categories for low-resolution face images [5]. FER+ adds multiple annotations for the same images, including contempt, unknown, and not-a-face responses [6]. Barsoum et al. already compare hard-label and distribution-based training methods. We build on that established learning principle by evaluating selective vote disagreement under matched supervision and isolated operational roles. We retain seven model outputs for compatibility with the prototype while accounting for all ten annotation categories in evaluation; our scores are not a replication of the original eight-expression FER+ benchmark.

Uncertain annotations have motivated FER-specific methods such as Self-Cure Network, which uses weighting and relabeling [7]. Our comparison instead changes the target representation of the same supported votes. Swin Transformer supplies hierarchical shifted-window features [8], and channel recalibration follows the established squeeze-and-excitation principle [9]. The architecture is held fixed so that supervision is the experimental variable. Vats and Chadha have also described Swin with squeeze-and-excitation for FER (unpublished preprint, 2023) [10].

More recent distribution-learning approaches exploit additional structure. Le et al. construct uncertainty-aware target distributions using neighbors in valence-arousal space [11]. MIDAS extends mixup to soft-labeled facial-expression videos [12]. Our study neither implements those mechanisms nor compares against their reported benchmark scores. It uses observed static-image votes directly to examine a matched supervision contrast and its selective outcomes.

Crowd-Calibrator connects annotation disagreement with calibration and selective prediction in subjective language tasks [13]. Our contribution is not that conceptual connection; it is the matched static-image target comparison and the fixed-set, label-reference and omitted-vocabulary diagnostics in this seven-output FER setting. Its language-task scores are not comparable FER benchmarks.

Selective classification examines the trade-off between rejection and error [14]; SelectiveNet learns a rejector jointly with prediction [15]. We use the simpler maximum-probability score to isolate the supervision comparison. This does not establish superiority over learned selectors. Softmax scores need not be calibrated probabilities [16]; an empirically fitted error target is not a guaranteed risk bound. Our reporting separates same-coverage ranking comparisons, calibration-fitted operating points, and disagreement inherent in the finite crowd annotations.

A closely related unpublished preprint by Wang, Li, and Wang combines vote-count modeling, predicted disagreement, and rejection for dynamic expressions (2026) [17]. Our static-image study addresses simpler target-representation controls and vocabulary diagnostics; it does not benchmark or claim priority over that framework.

3 Methods

3.1 Data provenance and isolated roles

FER2013 contains 28,709 Training, 3,589 PublicTest, and 3,589 PrivateTest rows. The CSV was acquired on 20 September 2026 from Kaggle mirror `deads skull17/fer2013`,

Table 1 Operational roles after exact-pixel isolation. Counts are images, not people

Source and role	Images	Use
Training	28,709	Optimization
PublicTest: selection	1,667	Checkpoint choice
PublicTest: calibration	1,642	Threshold choice
PrivateTest: primary	3,275	Main evaluation
PrivateTest: full	3,589	Prediction archive

Table 2 Adaptive experimental phases on the same source; every phase uses seeds 17, 42, 89 and selected/fixed-epoch evaluation

Phase	Target conditions	Runs	Prior outcomes known
1	Hard, Soft	6	Historical results
2	Uniform, Tie-hard	6	Phase 1
3	Tie-Uniform	3	Phases 1 and 2

All phases retain the same data roles and matched within-seed optimization. These are not independent-source replications.

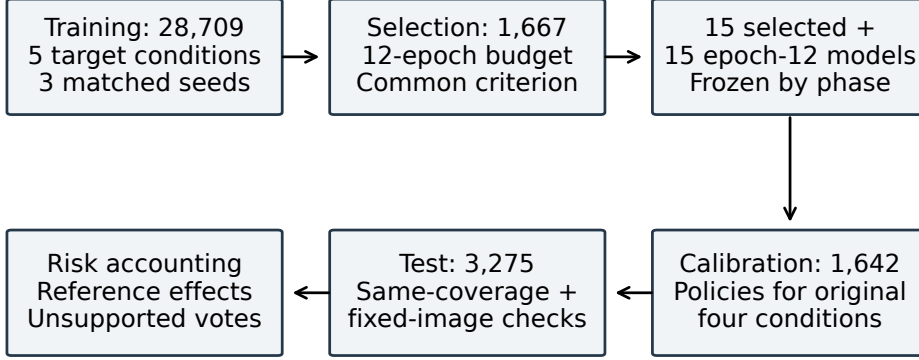
version 1, rather than an official maintainer distribution. FER+ annotations were obtained from the official repository and joined using original row identifiers and partition checks. Online Resource 1 records immutable sources, hashes, and row-level manifests. Images are not redistributed; a mirror’s advertised license does not independently establish rights to its underlying photographs.

The join follows the official FER+ conversion script’s row indexing, not its potentially blank or duplicated output filenames. A separate raw-source audit checks all 35,887 original emotion IDs, pixel byte hashes, and partition positions against the retained manifest. Named vote columns are reordered to the canonical model classes. These checks establish internal consistency with the acquired archive, not independent authenticity or correctness of every annotation.

Exact pixel hashes define the operational roles in Table 1 and Fig. 1. We keep the Training rows unchanged. After removing 280 PublicTest rows matching Training pixels, the remaining pixel clusters are assigned to checkpoint selection or calibration by a fixed hash rule, without consulting labels or outcomes. The primary test population removes 314 PrivateTest rows whose pixels occur in Training or any PublicTest row. No exact pixels are shared across the four roles. This is not verified person-disjoint or near-duplicate-disjoint evaluation. Full PrivateTest predictions are retained for audit; all main follow-up comparisons use the isolated primary population.

3.2 Matched hard and soft supervision

Let q_{ic} be the fraction of annotations for image i assigned to category c , normalized over all ten categories. The first seven correspond to the supported expression outputs. Define supported mass $w_i = \sum_{c=1}^7 q_{ic}$ and $r_{ic} = q_{ic}/w_i$ when $w_i > 0$. For model



Exact-pixel isolation; adaptive phases; not an independent dataset

Fig. 1 Offline experimental flow. Calibration transfer analyses retain the original four conditions; Tie-Uniform is a targeted secondary control. Checkpoint selection does not use calibration; all phases are adaptive rather than independent confirmation

probabilities p_{ic} , the two per-image objectives are

$$\ell_i^{\text{hard}} = -w_i \log p_{ih_i}, \quad h_i = \arg \max_{c \leq 7} r_{ic}, \quad (1)$$

$$\ell_i^{\text{soft}} = -w_i \sum_{c=1}^7 r_{ic} \log p_{ic} = - \sum_{c=1}^7 q_{ic} \log p_{ic}. \quad (2)$$

All conditions use the same images and weights. The 152 Training rows with zero supported mass contribute zero supervised loss rather than an invented category. Among positive-mass rows, 1,360 have tied maximum votes; the original hard condition resolves ties by fixed class order. Losses are averaged over batch size, not divided by total supported mass.

Two controls address what the original contrast cannot isolate. The dominant-confidence-preserving uniform-minority condition (Uniform) retains $m_i = \max_c r_{ic}$ at the canonical dominant class h_i , but assigns $(1-m_i)/6$ to each other class. The tie-aware hard condition (Tie-hard) assigns $1/|T_i|$ to each member of $T_i = \{c : q_{ic} = \max_{k \leq 7} q_{ik}\}$ and zero otherwise. Both minimize $-w_i \sum_c t_{ic} \log p_{ic}$. Uniform tests detailed minority allocation beyond image-dependent softening; it is not constant label smoothing and retains the canonical convention for tied maxima. Tie-hard is the expected loss under uniform random resolution among tied maxima, without additional random draws. It equals Hard on every unique-maximum image. These comparisons do not test discarding ambiguous images, manual relabeling, or training an expanded output vocabulary.

A third control, Tie-Uniform, preserves every tied maximum. With $k_i = |T_i|$, its target is

$$t_{ic}^{\text{TU}} = \begin{cases} m_i, & c \in T_i, \\ (1 - k_i m_i)/(7 - k_i), & c \notin T_i, k_i < 7. \end{cases} \quad (3)$$

When $k_i = 7$, all targets are $1/7$. Zero-supported-mass rows retain zero loss. This equals Uniform on unique maxima but avoids privileging one tied category. It preserves dominant confidence, not entropy or aggregate class marginals. Three additional matched runs test this secondary representation contrast, not a uniquely isolated mechanism.

3.3 Architecture, preprocessing, and optimization

All conditions use torchvision Swin-T with ImageNet-1K V1 initialization [8, 18, 19]. Its pooled feature $z \in \mathbb{R}^{768}$ is gated once, after the backbone:

$$\tilde{z} = z \odot \sigma(W_2 \text{ReLU}(W_1 z)), \quad W_1 \in \mathbb{R}^{48 \times 768}, \quad W_2 \in \mathbb{R}^{768 \times 48}. \quad (4)$$

The two gate layers omit biases; a biased linear layer produces seven logits. The gate adds 73,728 parameters. It is held fixed across the comparison, so the experiment cannot establish that gating is necessary or superior to another head. A dimensioned head schematic and an illustrative composition interface are supplied in Online Resource 1.

Images are converted to grayscale, replicated across three channels, resized to 224×224 using bilinear interpolation, and normalized with mean and standard deviation 0.5. The canonical output order is angry, disgust, fear, happy, neutral, sad, and surprise. Horizontal flips are the only augmentation; they are keyed deterministically to seed, epoch, and original row. Seeds 17, 42, and 89 produce three matched groups of five conditions with identical within-seed initialization, sample order, augmentation, architecture, and budget. After model construction, the classifier is reset with Torch seed $1000 + \text{seed}$. Initial-state hashes and configuration comparisons verify the match between original and control runs.

All backbone and head parameters are optimized for 12 epochs with AdamW [20], batch size 32, learning rate 10^{-4} , weight decay 10^{-4} , and a cosine scheduler with $T_{\max} = 12$ and $\eta_{\min} = 10^{-6}$, stepped after each epoch. AdamW uses $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$. Training uses BF16 autocast on an NVIDIA RTX 3070 Laptop GPU with 8 GB memory, PyTorch 2.8.0+cu128, and torchvision 0.23.0+cu128. Deterministic algorithms and seeded data loaders control this execution, not arbitrary cross-platform numerical behavior. The fixed budget is not a convergence or hyperparameter-optimality claim.

Checkpoint selection minimizes mean squared distance from the full ten-category vote distribution on the selection subset, with the seven predicted probabilities padded by three zeros. Earlier epochs win ties. This common criterion avoids selecting one condition by original-label F1. Each training phase is frozen before its calibration and test inference; original hard/soft results were known when the control protocol was written. Selected and fixed epoch-12 checkpoints are evaluated for all five conditions. The latter removes selection as a stopping decision but does not reconstruct selection by

an alternative metric, because all historical intermediate weights were not retained. Both sets of weights, logs, and probabilities are released. For fixed annotations, the squared distance differs from expected multiclass Brier loss against one annotation draw by $1 - \|q_i\|_2^2$, a model-independent constant. Thus, it provides a common probability-sensitive selection objective rather than privileging either categorical reference.

Nevertheless, the constrained quadratic and training objectives have different ideal targets. For one image with $w > 0$, weighted soft cross-entropy is minimized by $p_c = q_c/w$. Minimizing the ten-category squared distance under $\sum_{c \leq 7} p_c = 1$ instead gives $p_c = q_c + (1 - w)/7$, by the quadratic’s Lagrange multiplier. The solution is nonnegative and sums to one. These targets coincide at $w = 1$, but generally differ otherwise. The score remains legitimate; the mismatch motivates the fixed-epoch and supported-mass diagnostics rather than an assertion of criterion neutrality.

3.4 Disagreement, annotation floor, and classification metrics

For predicted category $\hat{c}_i = \arg \max p_i$, expected disagreement with one draw from the empirical annotations is $d_i = 1 - q_{i\hat{c}_i}$. It is not a claim about emotional truth. The best available seven-category prediction has floor $b_i = 1 - \max_{c \leq 7} q_{ic}$, giving

$$d_i = b_i + e_i, \quad e_i = \max_{c \leq 7} q_{ic} - q_{i\hat{c}_i} \geq 0. \quad (5)$$

The floor includes both disagreement among supported categories and votes outside the model’s vocabulary. We report the mean floor and excess e_i on the same accepted images as the total disagreement. They distinguish selecting less ambiguous images from choosing better categories, without treating the finite crowd distribution as a population oracle. Ten-category squared distribution distance additionally evaluates probability allocation, not just the winning category.

We further separate the floor using an accounting identity, not a new theorem:

$$d_i = \underbrace{1 - w_i}_{\text{unsupported mass}} + \underbrace{w_i - \max_{c \leq 7} q_{ic}}_{\text{supported ambiguity}} + e_i. \quad (6)$$

If an ideal model has $p_i = r_i$, its disagreement is $1 - w_i \max_c p_{ic}$. Maximum softmax probability (MSP) alone omits w_i : an MSP of 0.9 implies disagreement 0.1 when $w_i = 1$, but 0.55 when $w_i = 0.5$. This is a hypothetical consequence of the definitions, not evidence that omitted mass alone caused the observed transfer failure. Annotation-derived w_i is unavailable for a new runtime image and is never used as a deployable selector here.

We report $w = 0$, $0 < w < 1$, and $w = 1$ strata. Restricting each model’s global 80% accepted set differs from separately selecting 80% within each stratum; both are labeled. For positive-mass images, $1 - r_{i\hat{c}_i}$ is additionally reported as supported-only disagreement. This conditional endpoint changes the reference population of votes and is not a replacement for complete-vote risk or an expanded-vocabulary experiment.

Weighted and macro F1 against original FER2013 labels are secondary reference-specific diagnostics [21]. Weighted F1 is $\sum_c n_c F1_c / N$; macro F1 averages the seven $F1_c$

values equally, assigning zero to undefined class scores. Crowd-majority weighted F1 uses only images with more than half of all votes assigned to one supported category; its denominator is explicitly reported. It is not compared numerically with original-label F1 as if both were the same task. Full-vote outcomes retain images lacking a majority.

3.5 Selection, calibration, and uncertainty

The primary follow-up endpoint is Soft-minus-Uniform mean vote disagreement at 80% common test coverage. The original Soft-minus-Hard contrast is retained, with Tie-hard and fixed-epoch contrasts reported as sensitivity analyses. Within each fitted model, images are ranked by MSP; exactly $\lceil 0.8N \rceil$ are retained, with original row order resolving score ties. Conditions retain the same count, not necessarily the same images. This is a ranking evaluation, not an operational threshold fitted on test labels. Coverage values 20%, 50%, and 100% and all-prefix curves are secondary descriptions.

For deployment-style evaluation, thresholds are chosen on the separate calibration subset to maximize retained count subject to empirical mean vote disagreement no greater than 10%, 20%, or 30%. At least 100 calibration images must be accepted; score ties are never split. If no threshold qualifies, the policy abstains on all images. Each threshold is frozen and applied unchanged to the primary test population. Test risks may exceed their calibration targets; no confidence-bound guarantee is asserted.

For a conservative comparator, thresholds lie on a fixed 101-point grid from zero to one. At each threshold retaining $n \geq 100$ calibration rows, define

$$U = \min \left\{ 1, \bar{d} + \sqrt{\frac{\log(101/0.05)}{2n}} \right\}. \quad (7)$$

The policy maximizes coverage subject to U not exceeding the target, otherwise abstaining completely. Hoeffding’s inequality [22] and a union bound supply simultaneous per-model control over this fixed grid under independent, identically distributed bounded losses, a fixed fitted model, and an unchanged population. The same guarantee is not asserted for our reused, potentially dependent images or an unverified deployment population. This deliberately conservative comparator illustrates the coverage cost of a finite-sample margin; it is not a newly proposed risk-control method. Empty selections have undefined risk, not zero error.

For all empirical operating points, calibration-to-test changes in mean risk are partitioned into the three terms of Eq. 6, using each population’s accepted set. Unconditional population summaries are also reported. To examine within-calibration variability, 100 deterministic pixel-cluster half-splits fit each empirical threshold on one half and evaluate it on the disjoint half. Cluster assignments are shared across models. These overlapping resamples describe search optimism and sample variability, not independent replications or a causal apportionment of the test overshoot.

Each contrast is reported for all three paired seeds. A 5,000-replicate paired exact-pixel-cluster bootstrap shares sampled cluster multiplicities across compared models and reranks each replicate at 80% coverage. The percentile interval describes sample sensitivity conditional on fitted models, seeds, and observed annotations. It does not capture person-level dependence, annotation uncertainty, seed-population variability, or

Table 3 Selected-checkpoint risk at 80% coverage (2,620 of 3,275 images). Risk is mean $\pm$ sample SD; components are mean percentage points

Targets	Risk (%)	Unsupported	Ambiguity	Excess
Hard	26.30 ± 1.04	6.65	16.14	3.51
Tie-hard	25.82 ± 0.58	6.67	16.03	3.13
Uniform	24.06 ± 0.10	6.41	15.41	2.24
Tie-Uniform	24.27 ± 0.17	6.47	15.46	2.35
Soft	23.68 ± 0.07	6.55	15.01	2.13

Accepted identities differ; components provide accounting, not causal attribution.

prior hypothesis selection. Soft-minus-Tie-Uniform is a prespecified secondary contrast in the final adaptive phase. Only Soft-minus-Uniform is the primary follow-up contrast; other intervals are descriptive sensitivities without multiple-comparison significance claims. A zero-crossing interval does not establish equivalence.

3.6 Implementation and verification boundary

The sender-controlled local prototype is an implementation artifact, not the validation setting. It uses MediaPipe detection [23] and the shared image transform, class order and model head; invalid, stale or unapproved cues are suppressed. Detailed runtime, local-processing and transport boundaries are in the supplement. Code checks cover numerical metrics, checkpoint contracts and failure states. AI assistance (OpenAI Codex) was used for literature checking, experimental and analysis code, manuscript drafting, and technical verification. The authors retain responsibility for methods, results, source accuracy and final submission.

4 Results

4.1 Minority allocation, softening, and tied maxima

Table 3 compares all five target representations at the same retained count. Original-four-condition risk-coverage curves and selected/fixed-epoch contrast intervals are retained in the supplement; the direct primary difference curve appears below. The primary Soft-minus-Uniform difference is -0.38 pp (conditional 95% interval [-0.64, -0.07]). Paired differences for seeds 17, 42, and 89 are -0.35, -0.43, -0.35 pp. Full soft targets yielded lower selective disagreement than the dominant-confidence-preserving uniform-minority control under this matched protocol. The retained original Soft-minus-Hard contrast is -2.61 pp (conditional 95% interval [-2.98, -2.19]). Tie-hard minus Hard is -0.47 pp (conditional 95% interval [-0.80, -0.12]); Soft minus Tie-hard is -2.14 pp (conditional 95% interval [-2.49, -1.76]). These latter intervals are descriptive sensitivities, not additional primary hypothesis tests.

Table 4 Selected versus fixed epoch-12 complete-vote risk at 80% coverage (%); means across three paired seeds

Targets	Selected	Epoch 12
Hard	26.295	24.983
Tie-hard	25.824	24.576
Uniform	24.062	23.949
Tie-Uniform	24.272	23.939
Soft	23.684	23.632

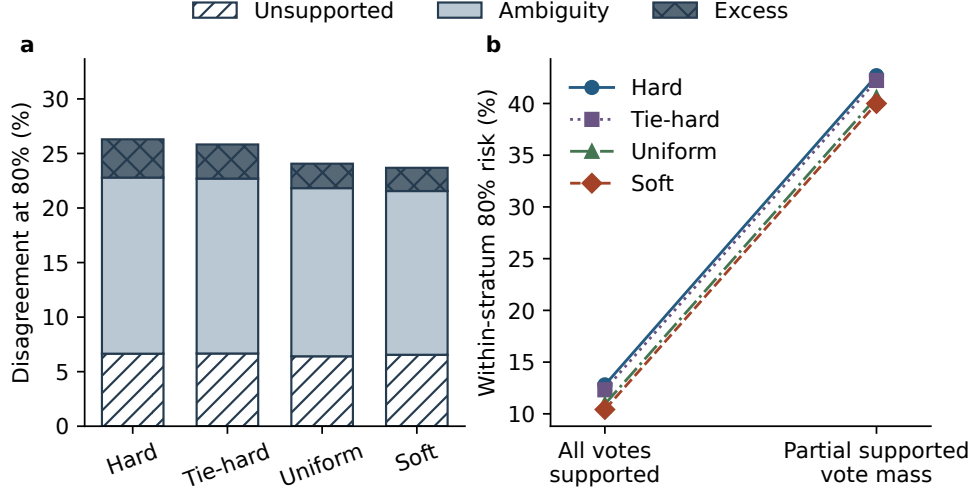


Fig. 2 Original four conditions: (a) three-way mean risk accounting on global 80% accepted sets. (b) Independent 80% selection within full- and partial-supported-mass strata; these populations and vertical scales differ from panel (a). Zero-supported-mass images necessarily have complete-vote risk one and are reported in Table 9 and Online Resource 1. Neither panel evaluates an expanded-vocabulary model

4.2 Selection sensitivity and label references

At fixed epoch 12, Soft minus Uniform is -0.32 pp (conditional 95% interval [-0.63, -0.05]), and Soft minus Hard is -1.35 pp (conditional 95% interval [-1.72, -1.05]). Table 4 keeps both evaluation states visible. This test changes the stopping decision, not the budget or training objective. Reference-specific classification outcomes are separated by both image population and label reference below.

4.3 Supported mass and the meaning of confidence

Figure 2 separates unsupported mass from ambiguity and excess on each accepted set. Its second panel changes the selection population explicitly rather than silently dropping unsupported votes. The primary population contains 1,732 fully supported, 1,527

Table 5 Cross-evaluation at 80% coverage: mean complete-vote disagreement (%) across seeds. Rows hold predictions fixed; columns hold accepted images fixed

State	Predictor	Uniform set	Soft set
selected	Uniform	24.062	24.037
selected	Soft	24.148	23.684
epoch12	Uniform	23.949	23.846
epoch12	Soft	24.037	23.632

Table 6 Selected-model weighted F1 with population and reference separated; means across three seeds

Targets	Original/all	Original/subset	Crowd/subset
Hard	0.5658	0.6580	0.8965
Tie-hard	0.5599	0.6572	0.9032
Uniform	0.5683	0.6642	0.9205
Tie-Uniform	0.5662	0.6619	0.9195
Soft	0.5722	0.6685	0.9270

All: 3,275 images; subset: the same 2,484 supported-majority images in the last two columns. No retraining or relabeling is performed.

partially supported, and 16 zero-supported-mass images. Within the fully supported stratum, 80% risk is 10.42% for Soft and 10.94% for Uniform. Within the partially supported stratum, 80% risk is 40.00% for Soft and 40.58% for Uniform. These descriptive comparisons address whether the primary contrast is confined to images with unsupported votes; they do not establish why a model assigned a particular score.

4.4 Fixed-image comparisons and reference effects

Table 5 holds the accepted image set fixed within each column. On Uniform-selected images, Soft minus Uniform is +0.085 pp; on Soft-selected images it is -0.352 pp. Within a column, annotations and the floor are identical, so the risk difference is also a category-choice excess difference on that image set. Holding the Uniform predictor fixed, changing from its own set to the Soft set changes risk by -0.025 pp; for the Soft predictor the corresponding change is -0.463 pp. Thus, the own-set contrast is not a consistent category-choice improvement on a fixed image set. Predictor and selector interact; these cross-evaluations are not a unique causal decomposition. The mean Jaccard overlap is 0.8408; individual intersections and overlap fractions are released.

In Table 6, the last two columns hold the image population fixed while changing the categorical reference. The first two hold the original-label reference fixed while changing the population. Their differences therefore disentangle two sources of the superficially large original-versus-crowd F1 contrast. The reference confusion matrix is supplied in the supplement; neither reference is treated as emotional truth.

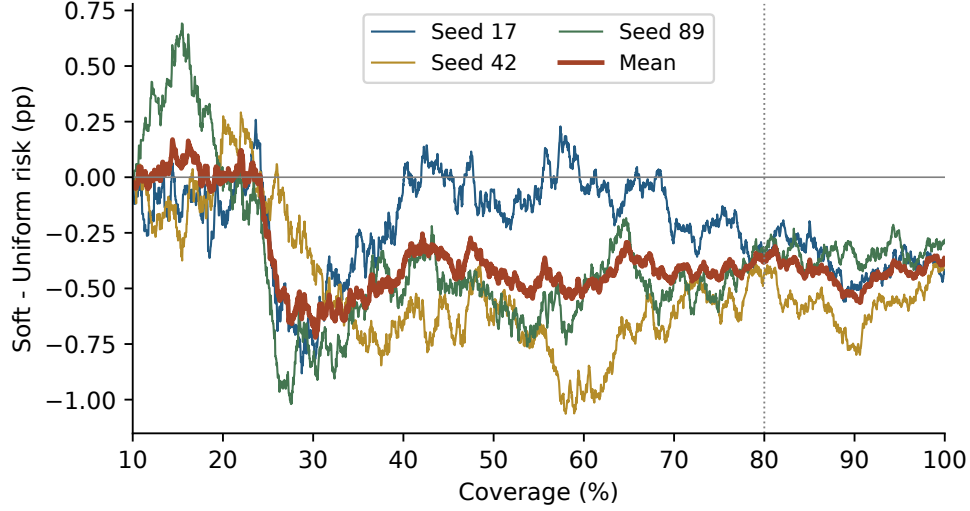


Fig. 3 Soft-minus-Uniform complete-vote disagreement versus common coverage. Thin curves are paired seeds; the thick curve is their mean. Curves use each model’s own ranking, not identical image sets. The vertical marker identifies the primary 80% endpoint; this descriptive curve has no simultaneous uncertainty band

Table 7 Secondary matched Tie-Uniform comparison at 80% coverage. Risks are percentages; differences and conditional 95% intervals are percentage points

State	Tie-Uniform	Soft	Difference	Interval
selected	24.272	23.684	-0.588	[-0.830, -0.281]
epoch12	23.939	23.632	-0.307	[-0.585, -0.024]

4.5 Tie-preserving dominant-confidence control

Table 7 reports both checkpoint states for the new secondary comparison. Selected: paired Soft-minus-Tie-Uniform differences are -0.378, -0.584, -0.802 pp for seeds 17, 42, and 89. The conditional interval lies below zero. Epoch12: paired Soft-minus-Tie-Uniform differences are -0.435, -0.221, -0.263 pp for seeds 17, 42, and 89. The conditional interval lies below zero. This secondary comparison removes canonical tie asymmetry from Uniform but still changes target entropy and class marginals relative to Soft. It cannot isolate a unique minority-category identity mechanism. Training-loss and selection-distance trajectories are supplied for every run in the supplement. Loss scales differ by target construction and are not directly comparable evidence of better fitting.

4.6 Retention of unsupported categories

Table 9 retains individual seed counts rather than only a mean ordering. Soft retains 11.00 of the 16 zero-supported-mass images on average, versus 7.00 for Uniform and

Table 8 Selected epochs under the common distribution-distance criterion

Targets	Seed 17	Seed 42	Seed 89
Hard	3	2	5
Tie-hard	4	4	5
Uniform	9	6	10
Tie-Uniform	12	8	7
Soft	9	10	10

Table 9 Accepted image counts at global 80% coverage, seeds 17, 42, 89 in that order. Population sizes: full 1,732; partial 1,527; zero 16

Targets	Full mass	Partial mass	Zero mass
Hard	1499, 1501, 1492	1115, 1112, 1122	6, 7, 6
Tie-hard	1519, 1494, 1508	1092, 1119, 1106	9, 7, 6
Uniform	1515, 1527, 1527	1098, 1087, 1085	7, 6, 8
Tie-Uniform	1529, 1522, 1520	1084, 1090, 1095	7, 8, 5
Soft	1527, 1527, 1533	1081, 1083, 1076	12, 10, 11

Each seed retains 2,620 images in total. Zero-mass images have disagreement one under every supported prediction.

6.67 for Tie-Uniform. Better aggregate risk therefore does not imply better rejection of wholly unsupported annotations. These small counts are descriptive, not an estimated deployment failure rate. Contempt, unknown and not-a-face vote-mass breakdowns are supplied separately rather than interpreted as one semantic category. Before confidence selection, calibration vote mass is 1.797% contempt, 3.666% unknown, and 0.262% not-a-face. Before confidence selection, test vote mass is 1.554% contempt, 5.667% unknown, and 0.656% not-a-face. These are vote fractions, not counts of images; the accepted-set breakdown in the supplement conditions on each ranking.

4.7 Calibration transfer and finite-sample diagnostics

Table 10 includes every target and empty-policy outcome. Figure 4 partitions the empirical policy’s accepted-population risk change. Across the original four conditions, 36 of 36 nonempty empirical test operating points exceed their nominal targets. These overlapping outcomes are not independent Bernoulli trials. At the 10% target, Hard has 38.35% mean coverage and 11.45% mean achieved risk. At the 10% target, Soft has 49.54% mean coverage and 13.01% mean achieved risk. Different achieved risks prevent interpreting this as greater coverage at equal risk.

Before confidence selection, the annotation floor is 23.70% in calibration and 26.67% in testing: a +2.97 pp difference. Unsupported mass changes from 5.72% to 7.88%; the remaining floor difference is supported-category ambiguity. Thus, these populations differ in annotation composition even before a model supplies a ranking.

Table 10 Calibration-fitted policies applied unchanged to primary testing. Entries are mean coverage / risk (%); coverage includes empty seeds, risk averages nonempty seeds

Target	Targets	Empirical	Fixed-grid bound
10%	Hard	38.35 / 11.45 (3/3)	0.00 / - (0/3)
10%	Tie-hard	39.17 / 11.93 (3/3)	0.00 / - (0/3)
10%	Uniform	45.99 / 12.22 (3/3)	0.00 / - (0/3)
10%	Soft	49.54 / 13.01 (3/3)	0.00 / - (0/3)
20%	Hard	68.58 / 22.28 (3/3)	47.03 / 14.55 (3/3)
20%	Tie-hard	71.04 / 22.80 (3/3)	48.05 / 14.61 (3/3)
20%	Uniform	76.20 / 22.80 (3/3)	57.49 / 16.11 (3/3)
20%	Soft	76.88 / 22.56 (3/3)	60.39 / 16.60 (3/3)
30%	Hard	98.06 / 32.39 (3/3)	82.01 / 26.92 (3/3)
30%	Tie-hard	99.47 / 32.37 (3/3)	85.41 / 27.54 (3/3)
30%	Uniform	99.94 / 31.46 (3/3)	88.41 / 27.11 (3/3)
30%	Soft	99.91 / 31.07 (3/3)	89.57 / 27.01 (3/3)

Parentheses give nonempty test seeds. An empty policy has undefined risk. Neither a nominal target nor the bound’s unverified independence/unchanged-population assumptions establish a deployment guarantee. Different achieved risks are not an equal-risk coverage comparison.

For Soft at the nominal 20% target, the accepted test-minus-calibration risk gap is +2.57 pp, comprising +1.98 pp unsupported mass, +0.78 pp supported ambiguity, and -0.19 pp excess. The figure and ledger supply the analogous accounting for every condition and target; this arithmetic does not identify independent causal contributions.

For Hard, the 20% calibration-half-split holdout-minus-fit risk gap averages +0.05 pp over 300 nonempty seed-split outcomes (descriptive 5th–95th percentiles -2.33 to +2.59 pp). For Tie-hard, the 20% calibration-half-split holdout-minus-fit risk gap averages +0.18 pp over 300 nonempty seed-split outcomes (descriptive 5th–95th percentiles -2.34 to +2.77 pp). For Uniform, the 20% calibration-half-split holdout-minus-fit risk gap averages +0.19 pp over 300 nonempty seed-split outcomes (descriptive 5th–95th percentiles -2.02 to +2.62 pp). For Soft, the 20% calibration-half-split holdout-minus-fit risk gap averages +0.17 pp over 300 nonempty seed-split outcomes (descriptive 5th–95th percentiles -2.00 to +2.43 pp). The complete target-specific resamples are released. These diagnostics reveal within-calibration variability and accepted-population differences, but cannot uniquely separate distribution shift, finite-sample fluctuation, and threshold-search effects. The fixed-grid comparator yields 24 nonempty test operating points, of which 0 exceed their nominal targets. It abstains completely at the 10% target for every condition and seed. Its observed conservatism therefore has a concrete coverage cost; these outcomes do not verify its population assumptions.

5 Discussion

The largest original Hard-to-Soft contrast primarily reflects changing the target representation rather than evidence that detailed minority identities alone are essential.

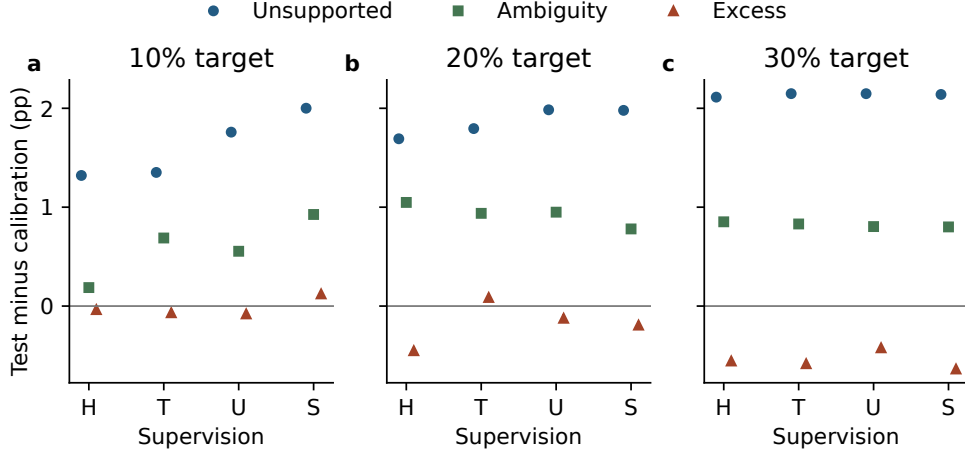


Fig. 4 Mean test-minus-calibration differences in unsupported mass, supported ambiguity, and excess error for empirical thresholds. Their sum is the risk gap, not a causal attribution. H: Hard; T: Tie-hard; U: Uniform; S: Soft. The three target panels share a vertical scale. Means use paired nonempty calibration/test selections

Uniform is a strong control, and Tie-Uniform removes its canonical tie asymmetry. Neither controls entropy or aggregate class marginals. The cross-evaluation matrix additionally shows that predictor comparisons depend on the accepted image set. An own-ranking advantage should not be promoted to a general claim of improved category choice.

The matched-reference table makes another reporting distinction concrete: changing image eligibility and changing labels both alter F1. Full-vote disagreement avoids silently excluding non-majority images, but remains conditional on the finite crowd panel. The zero-supported-mass retention counts are an adverse result that aggregate selected risk conceals. Contempt, unknown and not-a-face have different meanings; a seven-output model cannot resolve them simply by reporting high MSP. Tie-Uniform selected risk differs from Uniform by +0.210 pp. The residual own-ranking advantage persists under this tie-preserving control within the conditional interval. Because this adaptive check still uses the same source, three seeds and one model family, it is a representation sensitivity rather than independent replication. For scale, Uniform lowers selected-checkpoint risk relative to Hard by 2.23 pp, compared with 2.61 pp for Soft. The additional 0.38 pp at 2,620 retained images corresponds to about 9.9 fewer expected disagreements with an empirical annotation draw, not that many corrected images: accepted identities differ and losses are fractional. This arithmetic is not a causal fraction of improvement.

The distinction in Eq. 5 is important for interpretation. A lower selected vote-disagreement rate can reflect a lower annotation floor, a lower excess above that floor, or both. Reporting only total risk would conceal these different components. Conversely, the floor is not an immutable property of the person or even the image: it depends on this annotation panel, category vocabulary, and missing conversational context.

Extending the vocabulary could change the available-category floor, but that is not what the seven-output experiment tests.

The study addresses a narrower question than whether expression cues improve communication. A model may agree with annotators and still attach a category irrelevant to the sender’s intended meaning. Rejecting ambiguous examples may remove precisely the nuance the application aims to retain. Sender control gives an opportunity to suppress a cue, but its benefit and comprehensibility were not evaluated. The prototype is therefore an implementation artifact motivated by the measurement problem, not a validated intervention.

5.1 Limitations and scope of inference

The principal limitation is one reused image source. FER+ reannotates FER2013; it is not an independent image dataset. Earlier PrivateTest outcomes informed the adaptive follow-up phases. The six-control and later three-run Tie-Uniform protocols were each fixed before execution but after the preceding outcomes and reviews were known. Pixel isolation removes a specific overlap route, not identity leakage, near duplicates, dataset-selection bias, or independent-domain uncertainty. The findings are conditional within-dataset evidence rather than external confirmation.

Three paired seeds and a fixed 12-epoch budget constrain optimization claims. Fixed-epoch sensitivity does not establish convergence or insensitivity to all selection criteria. Grayscale preprocessing, small images, architecture, and supported-vote weighting can affect the contrasts. We do not compare a learned selector, another backbone, a second independently acquired vote-annotated image source, or an expanded-vocabulary model. Supported-mass stratification diagnoses the present vocabulary but does not replace such a model experiment. The controls test specific alternative explanations, not all mechanisms behind the observed differences. Favorable results would not establish algorithmic novelty; unfavorable results would not refute distribution learning generally.

Uniform preserves dominant confidence, not target entropy or all class marginals: distributing residual mass uniformly generally changes both. Its contrast with Soft tests these two specified target representations, not a unique causal effect of alternative-category identities or a proof that full distributions are necessary. Tie-hard addresses canonical ties only, not every alternative hard-label construction. The follow-up does not eliminate these remaining design dependencies.

FER images differ from detected faces during message composition. Lighting, pose, occlusion, accessories, camera quality, and detector failures may alter both predictions and rejection behavior. Demographic and cultural variation were not evaluated. The benchmark does not establish mobile efficiency, runtime error bounds, or communication benefit. No participant findings, survey ratings, or interpersonal conclusions are included.

The historical preprint described manual relabeling and reported scores whose curation records and checkpoints were not retained. Those scores are excluded from the new findings; their provenance is documented separately. The current experiment uses traceable public annotations, not an assertion that the historical manual process has been reproduced. Independent-image evaluation and an appropriately reviewed

communication study remain distinct future work, rather than evidence supplied by this revision.

6 Conclusion

This matched, adaptive within-source study finds that conclusions about crowd supervision depend on target softening, tie handling, accepted-image composition and reference labels. Detailed votes are not shown to be universally necessary, and lower own-set risk is not a uniform fixed-image prediction improvement. Unsupported-vote retention and calibration transfer failures constrain runtime interpretations. Releasing all selected/fixed-epoch weights, predictions and diagnostics makes these bounded findings auditable; independent-image and naturalistic messaging evaluation remain absent.

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Author contributions. Gunjan Kumar Mishra conceived the prototype and led manuscript preparation. Bijaya Ghimire contributed to implementation, experimental design, and manuscript revision. Badri Raj Lamichhane supervised the work, contributed to methodology review, interpretation, and manuscript revision, and serves as corresponding author. Alan Shah contributed to literature review and code testing. All four authors approved the authorship arrangement and declarations.

Ethics approval. This work reanalyzes existing benchmark images and annotations and performs software tests. No institutional ethics approval or exemption determination was obtained. The authors do not treat public availability as proof of institutional exemption.

Consent to participate. No participants were recruited and no new participant data were collected for this revision. Consent documentation for every photograph in the existing benchmark was not independently verified; public availability is not presented as proof of such consent.

Consent for publication. No identifiable facial photograph is reproduced. Interface illustrations contain synthetic text and no participant images.

Data availability. Acquisition records for FER2013 [5] and official FER+ annotations [6], immutable hashes, role assignments, and sample-level model probabilities are supplied in Online Resource 1. Images are not redistributed. The full ten-category annotations are retained for auditing the reported metrics.

Code availability. Online Resource 1 supplies experimental, analysis, and application code, tests, environment records, predictions, and package hashes. Online Resources 2–6 supply selected and epoch-12 inference checkpoints for Hard, Soft, Uniform, Tie-hard, and Tie-Uniform, respectively. The current code and evidence repository is <https://github.com/mgunjan67/crowd-supervision-selective-fer>, with versioned archives and inference checkpoints at <https://github.com/mgunjan67/crowd-supervision-selective-fer/releases/tag/v1.1.0>. The earlier project repository

remains historical provenance. Automated artifact regeneration is an internal verification, not an external replication.

Preprint statement. An earlier version appeared on TechRxiv as “Typing with Emotions: Facial Expression Recognition and Embedding in Text Messages,” version 1, <https://doi.org/10.36227/techrxiv.175416003.30236370/v1>. This revision excludes participant evidence and unverifiable historical performance claims, corrects implementation descriptions, and supplies newly documented experiments. It does not claim to reproduce the historical curation procedure.

Supplementary information. Online Resource 1: code, all three adaptive protocols, acquisition and split manifests, per-seed predictions and results, supplementary diagnostics, and historical provenance. Online Resources 2–6: all 30 selected and fixed-epoch inference checkpoints, grouped by supervision condition, with checksum manifests.

References

- [1] Wang, S.-M., Lee, C.-H.S., Lo, Y.-C., Huang, T.-H., Ku, L.-W.: Sensing emotions in text messages: An application and deployment study of EmotionPush. In: Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: System Demonstrations, pp. 141–145. The COLING 2016 Organizing Committee, Osaka, Japan (2016). <https://aclanthology.org/C16-2030/>
- [2] Chong, L., Jin, M., He, Y.: EmoChat: Bringing multimodal emotion detection to mobile conversation. In: 2019 5th International Conference on Big Data Computing and Communications (BIGCOM), pp. 213–221 (2019). <https://doi.org/10.1109/BIGCOM.2019.00037>
- [3] Liu, L., Guta, D., Niu, S.: Integration of user emotion and self-awareness in text messaging. In: 2019 IEEE 5th International Conference on Collaboration and Internet Computing (CIC), pp. 178–183 (2019). <https://doi.org/10.1109/CIC48465.2019.00030>
- [4] Barrett, L.F., Adolphs, R., Marsella, S., Martinez, A.M., Pollak, S.D.: Emotional expressions reconsidered: Challenges to inferring emotion from human facial movements. *Psychological Science in the Public Interest* **20**(1), 1–68 (2019) <https://doi.org/10.1177/1529100619832930>
- [5] Goodfellow, I.J., Erhan, D., Carrier, P.L., Courville, A., Mirza, M., Hamner, B., Cukierski, W., Tang, Y., Thaler, D., Lee, D.-H., Zhou, Y., Ramaiah, C., Feng, F., Li, R., Wang, X., Athanasakis, D., Shawe-Taylor, J., Milakov, M., Park, J., Ionescu, R., Popescu, M., Grozea, C., Bergstra, J., Xie, J., Romaszko, L., Xu, B., Chuang, Z., Bengio, Y.: Challenges in representation learning: A report on three machine learning contests. In: Lee, M., Hirose, A., Hou, Z.-G., Kil, R.M. (eds.) *Neural Information Processing. Lecture Notes in Computer Science*, vol. 8228, pp. 117–124. Springer, Berlin, Heidelberg (2013). https://doi.org/10.1007/978-3-642-42051-1_16

- [6] Barsoum, E., Zhang, C., Canton Ferrer, C., Zhang, Z.: Training deep networks for facial expression recognition with crowd-sourced label distribution. In: Proceedings of the 18th ACM International Conference on Multimodal Interaction, pp. 279–283. Association for Computing Machinery, New York, NY, USA (2016). <https://doi.org/10.1145/2993148.2993165>
- [7] Wang, K., Peng, X., Yang, J., Lu, S., Qiao, Y.: Suppressing uncertainties for large-scale facial expression recognition. In: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 6897–6906 (2020). <https://doi.org/10.1109/CVPR42600.2020.00693>
- [8] Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., Guo, B.: Swin transformer: Hierarchical vision transformer using shifted windows. In: 2021 IEEE/CVF International Conference on Computer Vision, pp. 9992–10002 (2021). <https://doi.org/10.1109/ICCV48922.2021.00986>
- [9] Hu, J., Shen, L., Sun, G.: Squeeze-and-excitation networks. In: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 7132–7141 (2018). <https://doi.org/10.1109/CVPR.2018.00745>
- [10] Vats, A., Chadha, A.: Facial Expression Recognition using Squeeze and Excitation-powered Swin Transformers. arXiv:2301.10906v7. Unpublished preprint (2023). <https://doi.org/10.48550/arXiv.2301.10906> . <https://arxiv.org/abs/2301.10906>
- [11] Le, N., Nguyen, K., Tran, Q., Tjiputra, E., Le, B., Nguyen, A.: Uncertainty-aware label distribution learning for facial expression recognition. In: 2023 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), pp. 6077–6086 (2023). <https://doi.org/10.1109/WACV56688.2023.00603>
- [12] Kawamura, R., Hayashi, H., Takemura, N., Nagahara, H.: MIDAS: Mixing ambiguous data with soft labels for dynamic facial expression recognition. In: 2024 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), pp. 6538–6548 (2024). <https://doi.org/10.1109/WACV57701.2024.00642>
- [13] Khurana, U., Nalisnick, E., Fokkens, A., Swayamdipta, S.: Crowd-Calibrator: Can annotator disagreement inform calibration in subjective tasks? In: Conference on Language Modeling (2024). <https://openreview.net/forum?id=VWWzO3ewMS>
- [14] Geifman, Y., El-Yaniv, R.: Selective classification for deep neural networks. In: Advances in Neural Information Processing Systems, vol. 30 (2017). <https://doi.org/10.48550/arXiv.1705.08500>
- [15] Geifman, Y., El-Yaniv, R.: SelectiveNet: A deep neural network with an integrated reject option. In: Proceedings of the 36th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 97, pp. 2151–2159 (2019). <https://proceedings.mlr.press/v97/geifman19a.html>

- [16] Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q.: On calibration of modern neural networks. In: Proceedings of the 34th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 70, pp. 1321–1330 (2017). <https://proceedings.mlr.press/v70/guo17a.html>
- [17] Wang, Y., Li, F.W.B., Wang, J.: Predicting Human Disagreement for Calibrated Dynamic Facial Expression Recognition. arXiv:2609.17130v1. Unpublished preprint (2026). <https://arxiv.org/abs/2609.17130>
- [18] Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., Fei-Fei, L.: ImageNet: A large-scale hierarchical image database. In: 2009 IEEE Conference on Computer Vision and Pattern Recognition, pp. 248–255 (2009). <https://doi.org/10.1109/CVPR.2009.5206848>
- [19] Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., Desmaison, A., Köpf, A., Yang, E., DeVito, Z., Raison, M., Tejani, A., Chilamkurthy, S., Steiner, B., Fang, L., Bai, J., Chintala, S.: PyTorch: An imperative style, high-performance deep learning library. In: Advances in Neural Information Processing Systems 32, pp. 8024–8035 (2019). <https://doi.org/10.48550/arXiv.1912.01703>
- [20] Loshchilov, I., Hutter, F.: Decoupled weight decay regularization. In: International Conference on Learning Representations (2019). <https://openreview.net/forum?id=Bkg6RiCqY7>
- [21] Sokolova, M., Lapalme, G.: A systematic analysis of performance measures for classification tasks. Information Processing & Management **45**(4), 427–437 (2009) <https://doi.org/10.1016/j.ipm.2009.03.002>
- [22] Hoeffding, W.: Probability inequalities for sums of bounded random variables. Journal of the American Statistical Association **58**(301), 13–30 (1963) <https://doi.org/10.1080/01621459.1963.10500830>
- [23] Lugaresi, C., Tang, J., Nash, H., McClanahan, C., Uboweja, E., Hays, M., Zhang, F., Chang, C.-L., Yong, M.G., Lee, J., Chang, W.-T., Hua, W., Georg, M., Grundmann, M.: MediaPipe: A framework for building perception pipelines. In: Third Workshop on Computer Vision for AR/VR at IEEE CVPR (2019). <https://doi.org/10.48550/arXiv.1906.08172>